$$= \left[\frac{2\sqrt{2}+2+\sqrt{2}}{2\sqrt{2}}\right]\left[\frac{2\sqrt{2}-2+\sqrt{2}}{2\sqrt{2}}\right] = \left[\frac{3\sqrt{2}+2}{2\sqrt{2}}\right]\left[\frac{3\sqrt{2}-2}{2\sqrt{2}}\right]$$

$$= \frac{18-4}{4\left(\sqrt{2}\right)^2} = \frac{14}{4\times 2} = \frac{7}{4}$$

(ii) $\tan^2 60° - 2\tan^2 45° - \cot^2 30° + 2\sin^2 30° + \frac{3}{4}\operatorname{cosec}^2 45°$

$$= \left(\sqrt{3}\right)^2 - 2(1)^2 - \left(\sqrt{3}\right)^2 + 2\left(\frac{1}{2}\right)^2 + \frac{3}{4}\left(\sqrt{2}\right)^2$$

$$= 3 - 2 - 3 + \frac{1}{2} + \frac{3}{2}$$

$$= -2 + \frac{4}{2} = -2 + 2 = 0$$

Note

(i) In a right angled triangle, if the angles are in the ratio $45° : 45° : 90°$, then the sides are in the ratio $1 : 1 : \sqrt{2}$.

(ii) Similarly, if the angles are in the ratio $30° : 60° : 90°$, then the sides are in the ratio $1 : \sqrt{3} : 2$.

(The two set squares in your geometry box is one of the best example for the above two types of triangles).

Exercise 6.2

1. Verify the following equalities:

   (i) $\sin^2 60° + \cos^2 60° = 1$

   (ii) $1 + \tan^2 30° = \sec^2 30°$

   (iii) $\cos 90° = 1 - 2\sin^2 45° = 2\cos^2 45° - 1$

   (iv) $\sin 30° \cos 60° + \cos 30° \sin 60° = \sin 90°$

2. Find the value of the following:

   (i) $\frac{\tan 45°}{\operatorname{cosec} 30°} + \frac{\sec 60°}{\cot 45°} - \frac{5\sin 90°}{2\cos 0°}$

   (ii) $(\sin 90° + \cos 60° + \cos 45°) \times (\sin 30° + \cos 0° - \cos 45°)$

   (iii) $\sin^2 30° - 2\cos^3 60° + 3\tan^4 45°$

3. Verify $\cos 3A = 4\cos^3 A - 3\cos A$, when $A = 30°$

4. Find the value of $8\sin 2x \cos 4x \sin 6x$, when $x = 15°$

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